



MANIPAL INSTITUTE OF TECHNOLOGY
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DNN CCM Final Presentation

Uncovering Causal Dynamics in Chaotic Systems: From Lorenz to Manifold Reconstruction

Leveraging State Space Analysis and Convergent Cross Mapping (CCM)

By

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What We Built?

A complete system that can reconstruct the full 3D Lorenz attractor from only the X component using:

1. 5 different autoencoder architectures (modular, switchable)
2. Hankel matrix delay embedding (based on Takens' theorem)
3. FNN regularization (ensures proper manifold structure)
4. Comprehensive comparison framework (evaluates all aspects)
5. Production-ready pipeline (easy to use, well-documented)

Innovation:

The combination of delay embedding + FNN regularization enables the "impossible" task of recovering full system dynamics from partial observations.

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